MIL-C-24643/57B
3 October 1986
SUPERSEDING
MIL-C-0024643/57A(SH)
8 November 1985
MIL-C-24643/57
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE AND SIZE LS2UW-42 (INCLUDING VARIATIONS LS2UWA-42 AND LS2UWS-42)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

This cable can be manufactured in three variations: LS2UW (unarmored), LS2UWA (armored), and LS2UWS (double overall shield).

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper conductor, tin coated, in accordance with ASTM B 286, size 26-7. Nominal diameter 0.020 inch.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene insulation. Nominal wall thickness, 0.010 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Telephone identification code applied by method 3.
Fourth - Two insulated conductors twisted together to form a pair having a 2 inch nominal lay.
Fifth - Forty-two pairs cabled together with a maximum lay of 16 times the overall diameter of the layer. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Fillers shall be employed to form a firm, well-rounded assembly.

(B) denotes changes.

AMSC N/A FSC 6145
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Sixth - A binder tape of polyester, type G in accordance with MIL-I-631, applied helically with overlap.
Seventh - A braided shield whose strands are AWG no. 34 tin coated copper. Minimum coverage of 85 percent.

Construction variant for type LS2UW-42

Eighth - A separator tape applied helically with overlap.
Ninth - Cross-linked polyolefin jacket. Nominal wall thickness 0.065 inch. Overall cable diameter 0.750 inch minimum, 0.790 inch maximum. Cable surface marking required.

Construction variant for type LS2UWA-42

Eighth - A separator tape applied helically with overlap.
Ninth - Cross-linked polyolefin jacket. Nominal wall thickness 0.065 inch. Overall cable diameter 0.800 inch minimum, 0.840 inch maximum.
Tenth - Braided metal armor with paint.

Construction variant for type LS2UWS-42

Eighth - Spirally wrapped 0.002 inch thick polyethylene terephthalate (mylar) separator tape with 25 percent minimum overlap.
Ninth - A braided shield whose strands are AWG no. 34 tin coated copper. Minimum coverage of 85 percent.
Tenth - Binder tape applied helically with overlap.
Eleventh - Cross-linked polyolefin jacket. Nominal wall thickness 0.065 inch. Overall cable diameter 0.800 inch minimum, 0.840 inch maximum. Cable surface marking required.

NOTES: 1. Variant LS2UW-42 military part number - M24643/57-01UO.
2. Variant LS2UWA-42 military part number - M24643/57-02AO.
3. Variant LS2UWS-42 military part number - M24643/57-03UD.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	45.19
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor.....	2000
Conductor to shield.....	1000
Shield to shield (LS2UWS-42 only), volts, minimum..	6
Megohms/1000 feet, minimum.....	1.0
Insulation resistance - megohms/1000 feet, minimum.....	500
Jacket flaws.....	No failure
Conductor and shield continuity.....	No failure

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group A:	
Visual and dimensional	No failure
Characteristic impedance - at 1 MHz, ohms	115 ± 15
Watertightness - leakage, in^3, maximum	1.0
Group B:	
Cold bending, cable - at minus 30 ± 2°C, over a 9-inch diameter mandrel	No damage
Bending endurance - at 25 ± 2°C, cycles, minimum	25
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Permanence of printing (jacket) - cycles, minimum	250
Heat distortion - percent of unaged, maximum	30
Cable filler removability	No failure
Shield - conformance to materials, construction and coverage	No failure
Armor - conformance to materials, construction and coverage	No failure
(B) Surface transfer impedance (type LS2UWS only)	
Milliohms per meter, maximum	700
EMP response time - ns/500 feet	60

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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

	Requirements
Aging and compatibility (cable) (125 ± 5°C).....	No failure
(B) Abrasion resistance (jacket) - scrapes, minimum.....	75
Acid gas equivalent - percent, maximum	
Jacket.....	2
Fillers.....	2
Insulation.....	18
Halogen content - percent, maximum	
Jacket.....	0.2
Fillers.....	0.2
Flame propagation (cable).....	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum.....	50
Elongation - percent of unaged, minimum.....	50
Smoke index, maximum	
Jacket.....	25
Fillers.....	—
(B) Insulation.....	45
Toxicity index, maximum	
Jacket.....	5
Fillers.....	5
Insulation.....	1.5
Durometer (jacket) - (type A) hardness, minimum.....	80
Weathering (jacket).....	No failure

UNIT ORDERING LENGTH: 1000 feet (nominal)

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Preparing activity:
Navy - SH
(Project 6145-1005-57)

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

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